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Equations Useful for Undergrad Physics Students

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Abstract

These are the Equations and Relations that i needed during studying in the Physics Departement as an undergrad student, I had problems memorizing most of these.

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CHAPTER 1

VECTORS

$$\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(A.C) - \vec{C}(A.B) \quad (1.1)$$

Line Element in different co-ordinates:-
in Cartesian co-ordinates:

$$dl^2 = dx^2 + dy^2 + dz^2 \quad (1.2)$$

in Cylindrical co-ordinates:

$$dl^2 = dr^2 + r^2 d\phi^2 + dz^2 \quad (1.3)$$

in spherical Co-ordinates

$$dl^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (1.4)$$

CHAPTER 2

TRIGONOMETRY

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta) \quad (2.1)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta) \quad (2.2)$$

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)] \quad (2.3)$$

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)] \quad (2.4)$$

$$\sin(\alpha)\sin(\beta) = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)] \quad (2.5)$$

$$\sin^2\alpha = \frac{1}{2}[1 - \cos 2\alpha] \quad (2.6)$$

$$\cos^2\alpha = \frac{1}{2}[1 + \cos 2\alpha] \quad (2.7)$$

$$2\sin x \cos x = \sin 2x \quad (2.8)$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (2.9)$$

$$\text{Cos}x = \frac{e^{ix} + e^{-ix}}{2} \quad (2.10)$$

$$\int_0^\pi \text{Sin}(\frac{n\pi}{a}x) \text{Sin}(\frac{n'\pi}{a}x) dx = \delta_{nn'} \quad (2.11)$$

Trigonometric Differentiation

$$\frac{d}{dx} \sin x = \cos x \quad (2.12)$$

$$\frac{d}{dx} \cos x = -\sin x \quad (2.13)$$

$$\frac{d}{dx} \tan x = \sec^2 x \quad (2.14)$$

$$\frac{d}{dx} \csc x = -\csc x \cot x \quad (2.15)$$

$$\frac{d}{dx} \sec x = \sec x \tan x \quad (2.16)$$

$$\frac{d}{dx} \cot x = -\csc^2 x \quad (2.17)$$

CHAPTER 3

HYPERBOLIC FUNCTIONS

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (3.1)$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (3.2)$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3.3)$$

$$\frac{\partial}{\partial x}(\cosh(x)) = \sinh(x) \quad (3.4)$$

$$\frac{\partial}{\partial x}(\sinh(x)) = \cosh(x) \quad (3.5)$$

$$\frac{\partial}{\partial x}(\tanh(x)) = \operatorname{sech}^2(x) \quad (3.6)$$

$$\frac{\partial}{\partial x}(\operatorname{sech}(x)) = -\operatorname{sech}(x)\tanh(x) \quad (3.7)$$

$$\frac{\partial}{\partial x}(\operatorname{csch}(x)) = -\operatorname{coth}(x)\operatorname{csch}(x) \quad (3.8)$$

$$\frac{\partial}{\partial x}(\operatorname{coth}(x)) = -\operatorname{csch}^2(x) \quad (3.9)$$

CHAPTER 4

DIFFERENTIATION

$$y = \frac{u}{v} \quad \longrightarrow \quad \frac{dy}{dx} = \frac{vu' - uv'}{v^2} \quad (4.1)$$

Trigonometric Differentiation

CHAPTER 5

EXPANTIONS

Theorem 1: Theorem Name

A theorem.

CHAPTER 6

FUNCTIONAL

$$\frac{\delta F}{\delta f(x)} = \lim_{\epsilon \rightarrow 0} \frac{F[f(x) + \epsilon \delta(x - x')] - F[f(x)]}{\epsilon} \quad (6.1)$$

$$J[x] = \int f^p(x) \phi(x) \, dx \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = P[f(x_o)]^{p-1} \phi(x_o) \quad (6.2)$$

$$H[f(x)] = \int g[f(x)] \, dx \quad \longrightarrow \quad \frac{\delta H}{\delta f(x_o)} = g'(f(x_o)) \quad (6.3)$$

$$J[f] = \int g(f'(y)) \, dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = -\frac{1}{dx} \left(\frac{d g(f')}{d f'(x)} \right) \quad (6.4)$$

$$J[f] = \int \left(\frac{\partial f}{\partial y} \right)^2 \, dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = -2 \frac{\partial^2 f}{\partial y^2} \quad (6.5)$$

$$G[f] = \int g(y, f) \, dy \quad \longrightarrow \quad \frac{\delta H}{\delta f(x_o)} = \frac{\partial g(x, f)}{\partial f(x)} \quad (6.6)$$

$$J[f] = \int g(y, f, f', f'') \, dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = \frac{\partial g}{\partial f} - \frac{d}{dx} \left(\frac{\partial g}{\partial f'(x)} \right) + \frac{d^2}{dx^2} \left(\frac{\partial g}{\partial f''} \right) \quad (6.7)$$

CHAPTER 7

MAXWELL EQUATION

(7.1)

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (7.2)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (7.3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} \quad (7.4)$$

CHAPTER 8

GAMMA FUNCTIONS

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt \quad (8.1)$$

$$\Gamma(n) = (n-1)! \quad (8.2)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad (8.3)$$

$$\int \ln(x) dx = x \ln(x) - x \quad \ln N! = N \ln N - N \quad (8.4)$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}} \quad (8.5)$$

$$\int_{-\infty}^{\infty} x^n e^{-\alpha x^2} = \begin{cases} \frac{1}{2} \left(\frac{1}{\alpha}\right)^{n+1} n! & n = 1, 2, 3, \dots \\ \frac{1}{2} \left(\frac{1}{\alpha}\right)^{\frac{2n+1}{2}} \Gamma\left(n + \frac{1}{2}\right) & n = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots \end{cases} \quad (8.6)$$

CHAPTER 9

ESSENTIAL SERIES

$$\sum_{n=0}^{N-1} e^{inx} = \frac{1 - e^{iNx}}{1 - e^{ix}} \quad (9.1)$$